BIG MODULAR — A0 / REVIEW ONLY

Complete pin-connected prototype circuit; NOT a validated power supply.
Default OFF. Dedicated 5 V or PD 9 V source only. No host-current negotiation.
Tentative simultaneous loads: 5 V input: 20 mA EACH rail; PD 9 V: 40 mA EACH rail.
No output ratings are warranted. User target currents, transient loads and noise limits remain unspecified.
PCB files: USB connector + mechanical holes ONLY. Absolutely NO tracks, vias or zones.
Read ../README.md before using or updating a PCB from schematic.

USB-C sink / permission / protected input

File: 01_input.kicad_sch

4.016 V intermediate supply / startup

File: 02_intermediate.kicad_sch

Bipolar preregulator: approximately +13.95 / −13.95 V

File: 03_bipolar.kicad_sch

6.52 V preregulator for clean +5 V

File: 04_five_prereg.kicad_sch

Low-noise postregulators / three output rails

File: 05_clean_rails.kicad_sch

16-pin Eurorack POWER OUT / harness only

File: 06_output.kicad_sch

NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

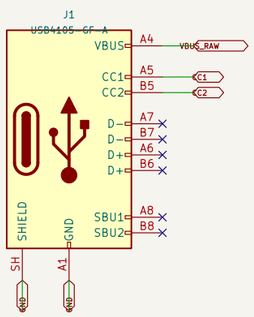
Sheet: /
File: big_modular.kicad_sch

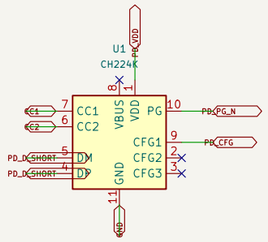
Title: Big Modular USB-C PD to +12V / −12V / +5V

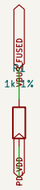
Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 1/7

USB-C sink / permission / protected input

9 V request only. No USB data. CH224K internal Rd: no parallel 5.1k resistors.
JP1 OPEN = OFF (default). Close ONLY for an approved dedicated source. PG is diagnostic, not a current-contract detector.
All bulk capacitance is behind the eFuse. No legacy-host enumeration or CC-current decoding is implemented.

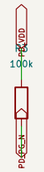




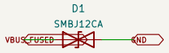




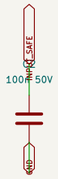


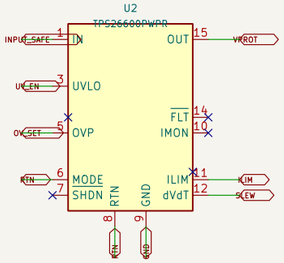


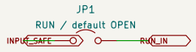






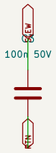




















NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

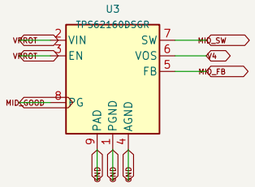
Sheet: /USB-C sink / permission / protected input/
File: 01_input.kicad_sch

Title: USB-C sink / permission / protected input

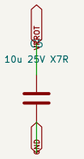
Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 2/7

4.016 V intermediate supply / startup

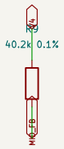
TPS62160: 3..17 V input, nominal 2.25 MHz DCS control, internal compensation.
2.2uH / 22uF LC follows datasheet baseline. Require effective output C within datasheet stability range.
PG releases converter enables only after intermediate rail rises. PWM frequency is not an audio-noise guarantee.

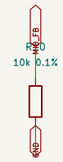


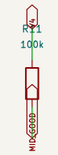


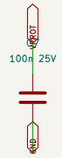












NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

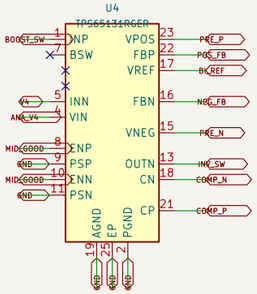
Sheet: /4.016 V intermediate supply / startup/
File: 02_intermediate.kicad_sch

Title: 4.016 V intermediate supply / startup

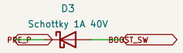
Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 3/7

Bipolar preregulator: approximately +13.95 / −13.95 V

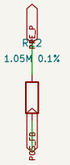
TPS65131: 2.7..5.5 V input; nominal 1.25 MHz. PSP/PSN LOW disables power-save mode.
Negative feedback references VREF, NOT ground. D3/D4 polarity follows boost/inverter energy paths.
BSW unused: positive preregulator can prebias through L/diode. No synchronized rail tracking promised.







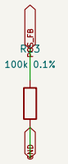


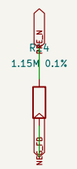








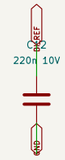


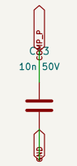




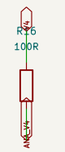


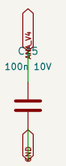


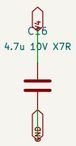














NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

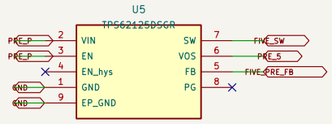
Sheet: /Bipolar preregulator: approximately +13.95 / −13.95 V/
File: 03_bipolar.kicad_sch

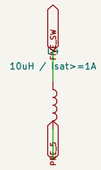
Title: Bipolar preregulator: approximately +13.95 / −13.95 V

Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 4/7

6.52 V preregulator for clean +5 V

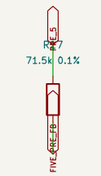
TPS62125 is explicitly used: TPS62160 cannot be substituted here (6 V recommended output limit).
10uH / 22uF combination is datasheet Table 2 recommendation for 5..10 V output.
DCS control operates UP TO 1 MHz and enters PFM at light load: strict always-MHz requirement is NOT met.

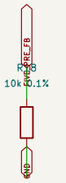


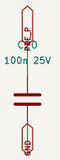












NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

Sheet: /6.52 V preregulator for clean +5 V/
File: 04_five_prereg.kicad_sch

Title: 6.52 V preregulator for clean +5 V

Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 5/7

Low-noise postregulators / three output rails

Positive LDO VFB=1.185 V; negative LDO VFB=-1.176 V (not the headline reference values).
Negative IN and EN connect to PRE_N; GND and exposed pad stay at 0 V. All capacitors here are non-polar MLCC.
10uF outputs must retain >=2.2uF under bias. 10nF NR/SS + feedforward: verify overshoot and startup on bench.
Diodes bypass reverse differential and clamp wrong-polarity excursions, NOT precise output overvoltage protection.

U6
TPS7A4901DGNR
PRE_P 8 IN
PRE_P 5 EN
P12_NR 6 NR/SS
3 NC
7 DNC
OUT 1 +12
FB 2 P12_FB
GND 4 GND
EP_GND 9 GND

C21 10u 25V X7R +12 GND
C22 10u 25V X7R +12 GND
R19 91.2k 0.1% +12 P12_FB
R20 10k 0.1% P12_FB GND
C23 10n 50V P12_NR GND
C24 10n 50V +12 P12_FB
D5 Schottky 1A 40V PRE_P +12
D6 Schottky 1A 40V +12 GND

U7
TPS7A3001DGNR
PRE_N 8 IN
PRE_N 5 EN
N12_NR 6 NR/SS
3 NC
7 DNC
OUT 1 +12
FB 2 N12_FB
GND 4 GND
EP_GND 9 GND

C25 10u 25V X7R PRE_N GND
C26 10u 25V X7R +12 GND
R21 92k 0.1% +12 N12_FB
R22 10k 0.1% N12_FB GND
C27 10n 50V N12_NR GND
C28 10n 50V +12 N12_FB
D7 Schottky 1A 40V +12 PRE_N
D8 Schottky 1A 40V GND +12

U8
TPS7A4901DGNR
PRE_5 8 IN
PRE_5 5 EN
P5_NR 6 NR/SS
3 NC
7 DNC
OUT 1 +5
FB 2 P5_FB
GND 4 GND
EP_GND 9 GND

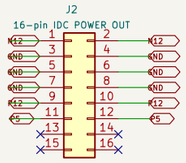
C29 10u 25V X7R +5 GND
C30 10u 25V X7R +5 GND
R23 32.2k 0.1% +5 P5_FB
R24 10k 0.1% P5_FB GND
C31 10n 50V P5_NR GND
C32 10n 50V +5 P5_FB
D9 Schottky 1A 40V PRE_5 +5
D10 Schottky 1A 40V +5 GND

NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS
Big Modular

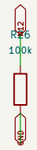
Sheet: /Low-noise postregulators / three output rails/		
File: 05_clean_rails.kicad_sch		
Title: Low-noise postregulators / three output rails		
Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 6/7

16-pin Eurorack POWER OUT / harness only

+5 V is on a separate STANDARD 16-pin interface, never added to the 10-pin standard.
J2 is an OFF-BOARD harness connector, not placed on either PCB. Exact crimp/IDC part and strain relief remain open.
RED STRIPE / pins 1 and 2 = −12 V. CV (13/14) and GATE (15/16) intentionally NC.
Do not trust key position alone. Continuity-test each cable; do not parallel this output with another supply.





























NOT FOR FABRICATION — UNROUTED PLACEMENT SCAFFOLDS

Big Modular

Sheet: /16-pin Eurorack POWER OUT / harness only/
File: 06_output.kicad_sch

Title: 16-pin Eurorack POWER OUT / harness only

Size: A3	Date: 2026-09-24	**Rev: A0 ENGINEERING PROTOT**
KiCad E.D.A. 10.0.6		Id: 7/7